



Integrating MCP into LangChain - Build Smarter Connected AI Systems

5th February 2026

WELCOME!

Get to know your Hosts!



Marina

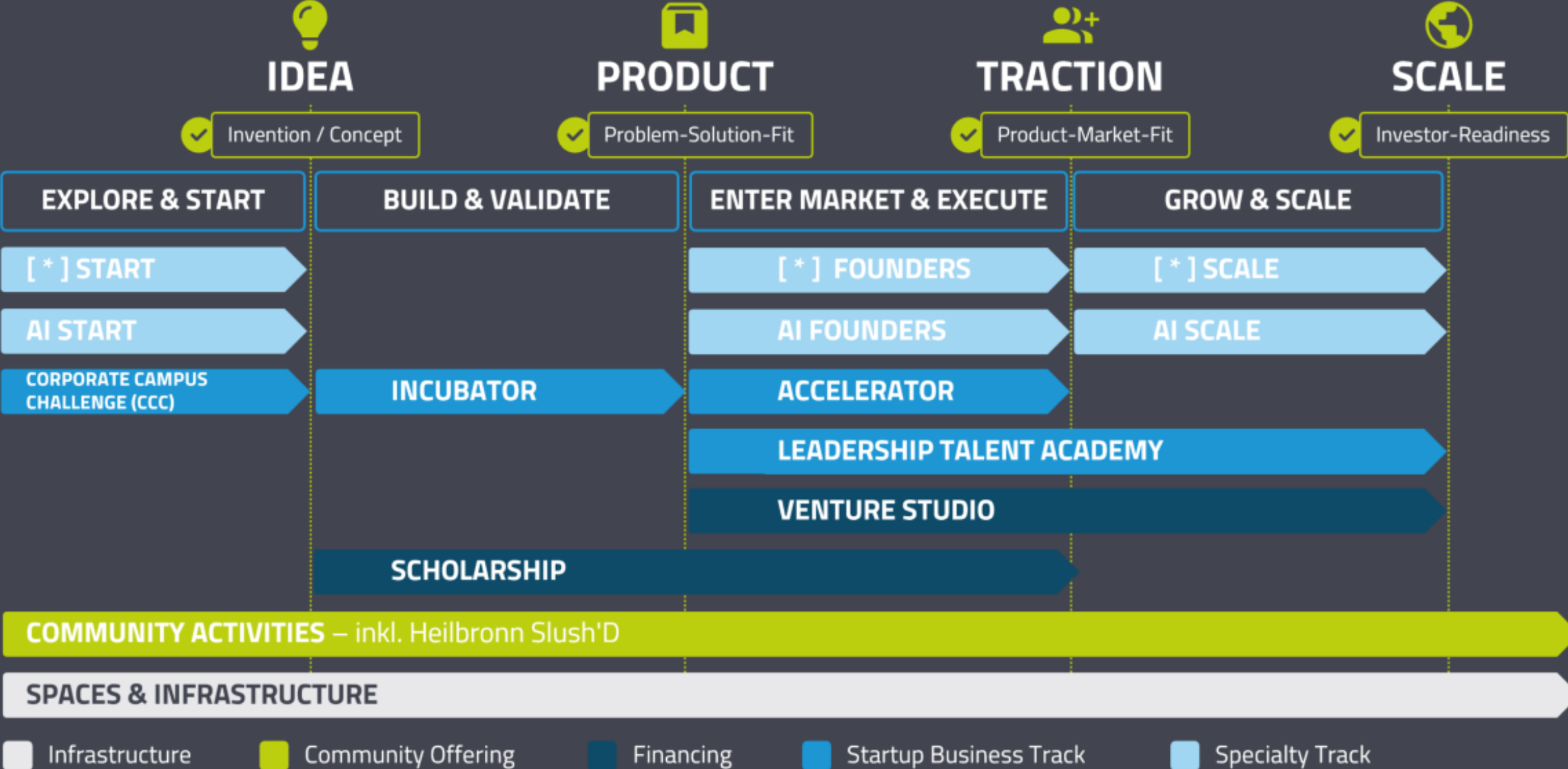
- AI Lead at Campus Founders since September 2024
- +9 years of experience in Data/AI
- + 12 years of experience in Academia
- Certified Principal Data Scientist, MSc in AI, Higher Diploma in Data Analytics, MBA
- +Co-Host a podcast ("AI Amigos Show", feel free to subscribe!)



Yacoub

- Founder and CEO of **PatentMuse**
- Founding member of **European Engineers United (EEU)**
- **AI and Software Engineer**, building production systems
- **42Hilebronn School**, systems and low-level programming

THE STARTUP JOURNEY



Successful Stories

There are many programs running at Campus Founders. Maybe you can be the next successful founder!





CAMPUS FOUNDERS

You Have Time to Apply!

- ❑ **AI Start:** Applications are NOW open. Deadline 9th February 2026
- ❑ **Incubator:** Get information. Application Process will open soon!
- ❑ **AI Founders:** Applications are NOW open. Deadline 1st March 2026



CAMPUS FOUNDERS

Let's Get Started

Sit and enjoy the ride!

What is LangChain?

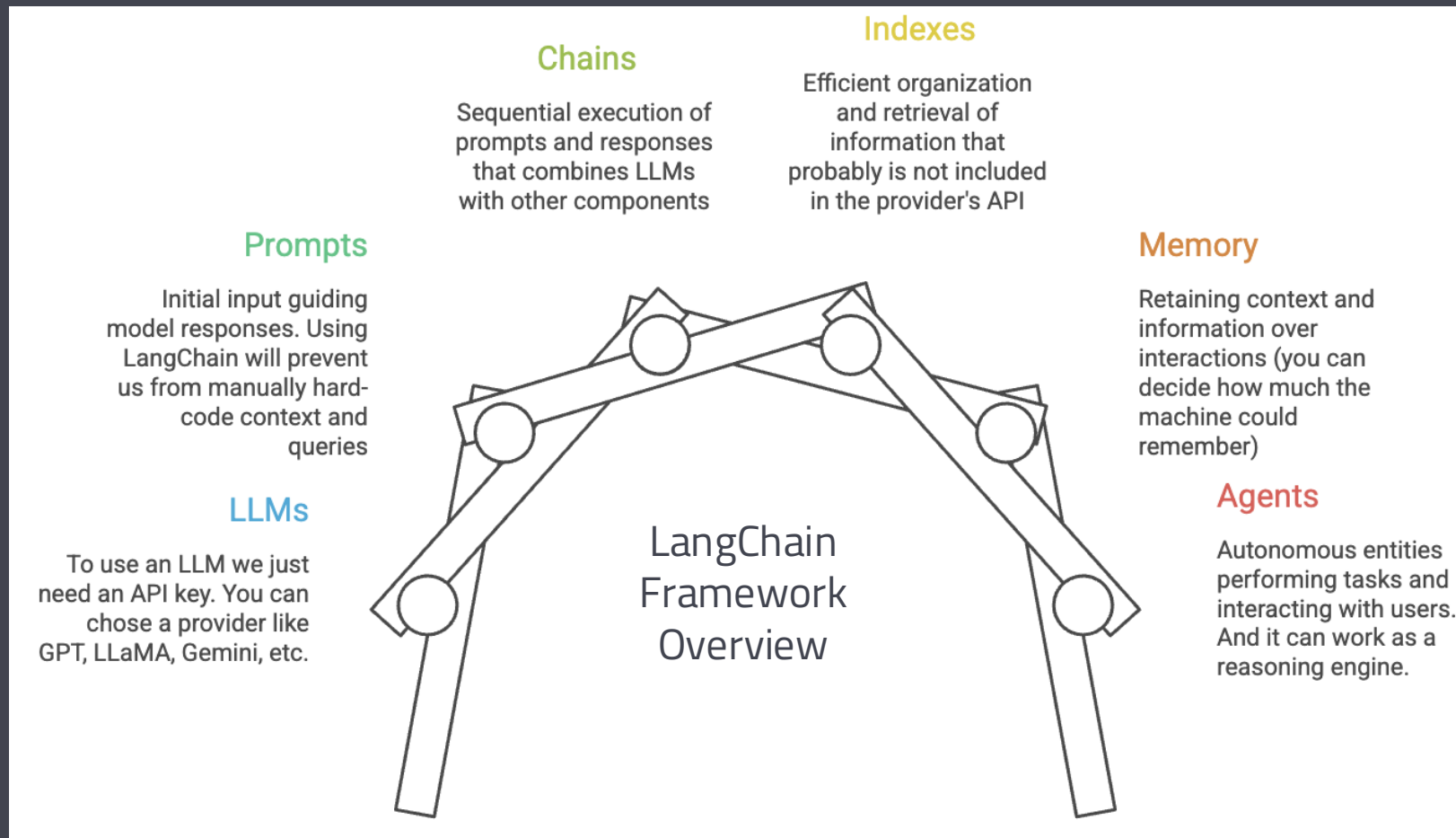
In simple words, LangChain is an open-source framework for the **development of applications** that use LLMs (IBM).

It can be defined as a **generic interface** for almost any LLM.

Once you use this kind of framework, you can **integrate** it with other components, such as data sources or software workflows.



Understanding the LangChain Framework



Key Components of the MCP

MCP HOST

The MCP host will connect you with the MCP server, and it could be in different formats, such as:

- Chat App
- Code assistant in your IDE.

This is the main AI application or environment (e.g., Claude Desktop, Windows).

MCP CLIENT

The MCP Client is a component within the host that initiates and maintains the connection to the server.

MCP SERVER

The MCP Server is a component within the host that initiates and maintains the connection to the server.

**Note: the MCP host can connect with multiple MCP servers.*

The MCP Host and Server will connect through the MCP Protocol.

The MCP server can connect with a DB, APIs, Data sources such as local files or code.

Let's Talk About Integration

MCP stands for Model Context Protocol, and it is an open standard designed to solve the "**integration chaos**" of connecting AI models to **external data** and tools.



Useful links

GitHub: <https://github.com/cheroualiyakoub/mcp-playground>

Colab:

<https://colab.research.google.com/drive/1gJ5X4ObXGDsnGAI81XwRQOAszcTbKheh?usp=sharing#scrollTo=ISnVUQeZzPSw>

API Key: `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`

Name: `OPENROUTE_API_KEY`



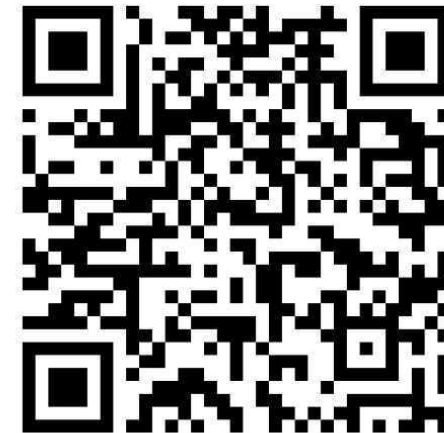
Marina Iantorno

AI Lead | Principal Data Scientist (DASCA) | AI
Strategy & Governance | LLMs & Synthetic Data |...



Yacoub Cherouali

Founder & CTO @ PatentMuse | AI ·
Semantic Search · Vector DBs



Let's Get in Touch!

THANK YOU